

Claude

Summary

Rather than re-deriving another theory-by-theory table (all three source models already did that separately), I organized this one by **gate** — for each of G0 through G10, what's the tripartite's own current status, which existing theory has already built the closest thing to a solution, exactly what transfers, and — critically, in its own column every time — what stays this program's unique burden regardless of the borrow.

Two things stand out from doing it this way rather than theory-first:

Gate G3 (the amplitude/tame-factor gate) has no donor at all, anywhere in the sixteen-theory survey across all three models. Every other gate has at least one rival program that's built something adjacent — G0 has Causal Fermion Systems, G8 has Hořava–Lifshitz's crossover shape and failure map, G10 has four independent gravity-as-residue lines. G3 has nothing. That's the single sharpest finding in the whole consolidation, and it's exactly the kind of thing a theory-organized table tends to bury.

The right-hand column repeats the same lesson eleven times in a row, which is itself the finding: every borrowed insight is a technique, a precedent, or a failure map — never the substrate, never Born statistics actually derived, never the reconciliation of the three theories' distinct roles. Section 5 states that as the program's real thesis, sharpened from "these three theories are correct" down to the much more defensible "every serious rival hits the same walls, and this program's falsifiable bet is that an arithmetic substrate gets furthest past them."

The compelling layer (Section 3) borrows z.md's strongest discipline — pairing every analogy with an explicit "breaks when" — rather than reproducing writeup_v1.1's fuller prose version, on the theory that the breakage tags are what keep this document from becoming exactly the kind of thing EBP exists to catch.

BM–IST–AS Tripartite Synthesis: Consolidated Gap-Filling Architecture

Drawing together v6.3.1, writeup v1.1, and the three-model QG-theory survey (deepseek, gemini, z) into one cohesive, coherent, robust, and compelling reference

Status: companion synthesis, not a new theory and not a supersession of v6.3.1. Everything here is read *through* v6.3.1's own constitution (EBP v2.1 — ideas enter free, promotion costs debt) and adds nothing that isn't already debt-tagged in one of the five source documents. Where this document states something as settled, it is settled only in the sense that a majority of independent passes converged on it — not in the sense that a gate has been closed.

1. What makes this cohesive, coherent, robust, and compelling — stated as four separate claims, because they are four separate properties

- **Cohesive:** BM, IST, and AS are not three theories stapled together. They are three answers to three different questions in one pipeline — what is the substrate (IST), how does its description change with scale (AS), what does it look like at the scale we actually inhabit (BM) — and the pipeline only holds together because each theory's most distinctive claims have been demoted to *targets* the pipeline must derive, not axioms it gets to assume. Section 2 restates this as one line.

- **Coherent:** every object lives in a typed space (state space (X) , measure space $(\mathcal{M}(X))$, theory space $(\mathcal{T})$, observable space $(\mathcal{O})$), and the single most important correction in this whole lineage — banning the ill-typed identity $(I_U = W^s(\Gamma_*))$ — is what actually makes that possible. Coherence here is a technical property, not a rhetorical one.
- **Robust:** the program has a symmetric Occam guillotine (kill conditions for the arithmetic layer, for AS, for BM, for E8, and for EBP itself), a pre-registered null-model roster, and an independent-reviewer requirement. It is robust in the specific sense that it is built to survive being wrong in a localized, reproducible way, not in the sense that it currently withstands scrutiny.
- **Compelling:** this is the one property that a table of theorems cannot supply by itself. Section 3 is the compelling layer, and it is compelling only because every analogy in it is paired with an explicit note on where it breaks — a discipline borrowed directly from the strongest of the five source documents, because an analogy without a breakage tag is exactly how a metaphor becomes an unearned axiom.

2. The architecture in one line, and the four objects that carry all the weight

\$(X,F) \rightarrow (\mathcal{C}_s) (\mathcal{M}(X),\Gamma_s,\mu_s) \rightarrow (\Phi,\mathcal{R}_s) \Gamma_\mu \rightarrow (\Pi) \psi,\rho,S,v \rightarrow \text{recovery} \} \text{QFT + GR} \$

Four objects do all the work, and everything else in the program is downstream of them:

- (X) — the candidate microscopic state space (IST's contribution, typed but not yet constructed);
- $(\mathcal{R}_s)$ — the candidate coarse-graining/RG flow (AS's contribution, a tool family, not a sacred equation);
- (μ_*) — the invariant measure whose IR projection must equal $(|\psi|^2)$ (the Born chain, B0–B3);
- (Π) — the observable projection that turns any of the above into something BM's guidance equation can act on.

Every gate in Section 4's table is a debt owed by one of these four objects.

3. The compelling layer: one spine, paired breakage tags

Non-technical readers get one continuous story; technical readers get the same story with the informal seed replaced by its typed object. Every entry carries a **breaks when** tag, because that tag is what keeps the analogy from quietly becoming a claim.

#	Everyday anchor	Technical anchor	Breaks when
1	A photograph is pixels, not continuous color — reality's bottom layer is discrete	Microscopic state space (X), finite information capacity (L)	The universe isn't shown to have a pixel grid — only that one is being tested as a hypothesis class
2	Postal codes: two addresses are "close" if they share a prefix, not if they're near on a map	(p)-adic ultrametric ($\ x-y\ _p$)	Physical space still needs to be Euclidean at human scale — the ultrametric governs <i>state space</i> , never physical distance directly

3	A river flows only where the bed is carved, not anywhere the water could geometrically go	Invariant set (I_U) / preperiodic locus of canonical height zero	The riverbed is carved by geology we understand; (I_U) 's carving mechanism is exactly what's unconstructed
4	Zooming a camera out blurs pixels into smooth shapes	Coarse-graining map $(\mathcal{C}_s)$, RG flow $(\mathfrak{R}_s)$	A camera's blur is a known, fixed algorithm; the substrate's actual $(\mathcal{C}_s)$ is a theorem target, not a filter anyone has written down
5	A ball rolling in honey — you don't track molecules, you get drag and jitter	Tracing out the bath: memory kernel $(K(t-s))$, colored noise $(\eta(t))$	This requires an actual tensor factorization or conditional-expectation structure to exist first — without it, "honey" is a metaphor, not a mechanism

6	Sheet music guiding dancers who always have a definite position	Wave function $(\psi=\sqrt{\rho},e^{iS/\hbar})$, guidance law $(v=\nabla S/m)$	The dancers' positions are given; here, both (ψ) and the guidance law are things the pipeline has to <i>produce</i> , not things it starts with
7	All liquids near their boiling point behave identically, regardless of which molecule	Universality, fixed point (Γ_*) , relevant/irrelevant directions	Universality across substrates has never been tested here — it is a cross-substrate theorem target, not yet a demonstrated fact
8	The waterline on an iceberg — where the visible part hands off to what's submerged	The crossover scale, where substrate hands off to continuum	The iceberg's waterline is directly measurable; the crossover scale is currently a symbol, not a computed number

This is the whole compelling layer stated honestly: eight analogies, eight explicit failure points. A reader who wants the version without the breakage tags can find it in writeup v1.1's Sections 6–14 — this table exists specifically so that version is never mistaken for the full story.

4. The centerpiece: gate-by-gate, where the rest of quantum gravity validly slots in

This is the table the synthesis exists to produce. It is organized by **gate**, not by donor theory, because the request was to fill the tripartite's own gaps — a theory-first table (which deepseek, gemini, and z each already produced separately) answers "what does LQG offer," while a gate-first table answers the sharper question: "for each specific thing this program still owes, is there an existing theory that has already built part of the answer, and what, even then, is still ours alone to finish."

Gate	What it demands	Tripartite's own status	Best-fit donor theory (net-valid insight)	What transfers	What remains the tripartite's unique burden

G0 — discrete/continuous typing consistency	No object may be silently treated as living in two incompatible spaces at once	Corrected in v6.3 (the $(I_U=W^s(\backslash\Gamma_*)$ ban); otherwise a standing discipline, not a solved gate	Causal Fermion Systems (Finster)	A published, theorem-backed instance of exactly this chain — discrete measure space + variational principle → Minkowski space + Einstein–Dirac equations, with the typing kept consistent throughout	CFS has no dynamics at all — no update law, no time, no Born mechanism. It proves the <i>typing discipline</i> is achievable, not that <i>this</i> substrate achieves it
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G1 — explicit microscopic map (F)	A concrete, constructed ((X,F)), not a schema	OPEN — Phase 1's central deliverable, not yet attempted	Wolfram-model hypergraph rewriting; causal-set sequential growth	Working code and methodology for building deterministic (Wolfram) or stochastic-growth (causal sets) update rules of comparable finite complexity	Both are the program's own <i>pre-designated null models</i> , not templates to imitate — their known failure points (no unique measure, stalled projection) are exactly this program's warning signs, and neither is arithmetic
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G2 — joint arithmetic compatibility (canonical height <i>and</i> symbolic/Markov structure on the <i>same</i> map)	One map (F) satisfying two independently-motivated arithmetic conditions simultaneously	UNTESTED — flagged as an open joint-satisfiability question	Arithmetic dynamics literature generally (Call–Silverman canonical heights; Bowen-type Markov partition theory)	Existing worked examples (e.g. algebraic doubling-type maps) where both structures are known to coexist, narrowing the search before new construction begins	No rival TOE candidate places number theory at the bottom at all — this test has no donor for the <i>substrate itself</i> , only for the mathematical toolkit used to check it
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G3 — amplitude-factor existence (the tame/Kronecker factor carrying $(U(1))$ phase)	A nontrivial zero-entropy factor of a genuinely mixing dynamics, surviving the $(L\rightarrow\infty)$ limit	The program's hardest and most distinctive gate — no rival theory has even opened it	<i>(none)</i> — genuinely unclaimed across the entire surveyed landscape	—	This is the single sharpest finding of the whole cross-model survey: every other discrete-substrate program either assumes a Hilbert space from the start or never needs a phase sector at all. Gate 3 is this program's alone to win or lose
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G4 — $(\exists \mu_*)$ with IR-sufficient regularity	A unique invariant measure, regular enough to carry finite Fisher information at observable scale	OPEN — construction not attempted	Causal Dynamical Triangulations (finite-size scaling, numerically produced spectral-dimension flow); tensor-network/holographic error-correction literature	A working numerical laboratory showing continuous macroscopic quantities <i>can</i> emerge from discrete building blocks, plus a rigorous framework (quantum error correction) for why some information stays hidden without violating unitarity	CDT's measure is a path-integral sum over geometries, not an invariant measure of a deterministic map — the actual object (μ_*) of $((X,F))$ has no external existence proof
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G5 — explicit observable projection ($\backslash P_i$)	A concrete, faithful map from microscopic/effective description to measurable quantities	OPEN, theorem target	AdS/CFT holography; Causal Fermion Systems' continuum-limit variational program	A worked example of an explicit boundary-to-bulk (or discrete-to-continuum) projection that has actually been carried through to concrete equations	Holography's boundary and bulk are both already well-understood quantum systems; this program's "boundary" is a bare arithmetic substrate — the projection has to do more work here than in any donor case
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G6 — Born identification, $(\Pi_{\rm IR} \mu = \psi ^2)$	The invariant measure's IR projection must equal the Born distribution, with a stated resolution-scale for any residual texture	OPEN — resolved <i>conceptually</i> in v6.3 by scale-typing (exact at IR projection, texture only below observational resolution); not yet computed	't Hooft's cellular-automaton interpretation (deterministic permutation → unitary evolution + Born-rule statistics via information loss/gauge equivalence)	A precedent that deterministic discrete dynamics <i>can</i>, in a worked toy setting, generate Born-rule statistics — proof of concept, not proof for this substrate	't Hooft's construction needs its own hand-picked information-loss /gauge-equivalence structure per model; nothing hands this program the <i>arithmetic</i> mechanism that would do the same job here
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G7 — effective dynamics ($\hbar \rightarrow 0$) Schrödinger/BM limit	The reduced Langevin/Kramers dynamics must have a controlled strong-friction limit landing on $(\gamma = \hbar \nabla^2 S/m)$	THEOREM TARGET, not attempted	Reginatto / Hall–Reginatto (Fisher-information derivation of the Schrödinger equation)	Mature, established mathematics for deriving Schrödinger dynamics from an exact uncertainty principle — the strongest single piece of "established mathematics" anywhere in this program's toolkit	Reginatto's derivation starts from an already-continuous configuration space with a postulated Fisher-information functional — it does not derive <i>that</i> starting point from a discrete substrate, which is exactly this program's actual burden
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G8 — guidance law and quantum potential (Q) recovered from the effective action	(Q) must fall out of the flow, not be inserted by hand	THEOREM TARGET	Hořava–Lifshitz gravity (anisotropic-scaling continuum effective theory with UV dimensional reduction)	A worked continuum shape for a UV-preferred-ordering theory crossing over to an IR-symmetric one — and, just as usefully, a documented graveyard of exactly how such crossovers fail (surviving Lorentz-violating operators, extra scalar modes)	HL's failure modes are a checklist, not a solution — this program still has to show its own crossover avoids every trap HL didn't
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G9 — composition, entanglement, Bell, no-signaling	Two weakly-coupled subsystems must yield $\mathcal{H}_{AB} \simeq \mathcal{H}_A \otimes \mathcal{H}_B$ (or an equivalent), with a double-counting audit against redundant nonlocality mechanisms	OPEN, and the double-counting risk is explicitly named	Tensor-network/holographic quantum error correction ("it-from-qubit")	A rigorous, citable foundation for treating locality as a <i>code property</i> — giving any future "protected/hidden sector" claim in this program an actual theorem class to borrow rather than assert	Borrowing the error-correction <i>formalism</i> is not the same as showing this substrate's own tame factor generates a code with the right protected structure — that map is still unbuilt
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G10 — QFT + Lorentz + gauge + GR recovery	The full relativistic/gravitational IR must fall out, not be assumed	RECOVERY PROGRAM, long-horizon	Jacobson / entropic gravity / Verlinde / holographic approaches (gravity as thermodynamic residue) — four independent lines converging on the same claim	Strong, independently-re plicated evidence that treating the metric sector as <i>downstream</i> of a more fundamental statistical structure is a defensible research strategy, not a fringe move	None of the four donor lines specifies what the fundamental structure actually <i>is</i> — they license the strategy, not the substrate
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Cross-cutting — spectral dimension flow ($d_s(k):2\rightarrow 4$)	A computed flow of effective dimension from UV to IR	UNVERIFIED, not computed for this substrate	Asymptotic Safety, LQG, causal sets, and CDT independently converge on ($d_s\approx 2$) in the UV	A five-way convergence — including one genuine numerical production line (CDT) to benchmark any future computation against	Convergence across other programs is evidence the target is a reasonable one to aim at; it is not evidence this substrate hits it
Cross-cutting — arrow of time / thermodynamic irreversibility	Irreversibility must be derived from the coarse-graining structure itself	TARGET, <code>needInitialCondition</code> debt open	Semiclassical/stochastic gravity (Einstein–Langevin framework, rigorously defined stress-energy noise kernel)	The gravity-side mathematics for how a derived stochastic source (once this program's own bath is constructed) would actually move an effective geometry	The noise kernel is a <i>toolkit</i>, not a source — it says nothing about why the substrate's own initial condition should have been special in the first place

5. The finding underneath the table: what none of the donor theories supplies, and why that's the actual thesis statement

Read down the right-most column of Section 4 and one pattern repeats without exception: every donor theory hands over a **technique, a precedent, or a failure map** — never the substrate itself, never the amplitude gate, never Born statistics derived rather than assumed. Stated as the four load-bearing claims no rival program shares:

1. **Determinism at the bottom.** Every surveyed rival except the Wolfram-model family is stochastic or sum-over-histories at the fundamental level. A genuinely deterministic (F) is this program's own commitment to prove, and the causal-set/CDT literature on Lorentz invariance suggests it may be the *harder* branch to reconcile with relativity, not the easier one.
2. **Arithmetic as the incumbent substrate.** No rival theory in the entire survey places number theory at the bottom. LQG and causal sets are geometric; Wolfram and 't Hooft are computational; string/matrix models are algebraic but not arithmetic in the specific sense this program means. The (p)-adic/adelic/height/Markov layer remains this program's distinctive, unshared wager.
3. **Born statistics from the substrate.** Typicality arguments exist (standard BM); growth-law measures exist (causal sets). Nothing in the surveyed landscape derives $(|\psi|^2)$ from a discrete deterministic dynamics. Gate G3/G6 together are unwon everywhere, not just here.
4. **The reconciliation itself.** Every donor theory, honestly imported, becomes exactly one of four things: a precedent (an existence proof that some version of the chain can work), a template (a technique for a specific named gate), a constraint (a failure map of traps to avoid), or a null model (a cheaper rival to be tested against and, ideally, beaten). None of them becomes a fifth theory added to the stack — and that discipline, more than any individual borrowed result, is what keeps this synthesis from collapsing back into theory collage.

The honest way to state the program's bet, after this survey, is not "BM, IST, and AS are correct and the rest of quantum gravity confirms it." It's narrower and more defensible: **every serious rival research program has independently run into the same handful of walls, and this program's specific, falsifiable claim is that an arithmetic substrate is the one that gets furthest past them.** The survey doesn't prove that claim. It does show the claim is not obviously wrong, and — just as importantly — it shows exactly which walls are shared (measure existence, the projection problem) versus which one belongs to this program alone (the amplitude gate, G3).

6. Supporting convergence, held at arm's length

Two cross-cutting observations from the survey are worth recording as *context*, not as evidence for this program specifically:

- **The shared graveyard validates the gate ordering.** Every discrete quantum-gravity program that has been tried stalls at the same two objects — an invariant/canonical measure, and the projection to observables (G4/G5 here). That convergence is why this program's gate order places those exactly where it does; it is not evidence this program will get past them.
 - **The gravity-as-residue genre has four independent lines of support** (Jacobson, entropic gravity, holography, Verlinde), all agreeing the metric sector is downstream of something more fundamental. This licenses G10's overall strategy — treat gravity as a recovery target, not a microscopic input — as a reasonable one to be attempting at all, without licensing any specific mechanism for how this program's own substrate would produce it.
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7. Provenance

This table consolidates: the twelve-theory survey and summary table in [deepseek.md](#); the four-pillar architectural mapping and Causal Fermion Systems addition in [gemini.md](#); the sixteen-group survey (through Hořava–Lifshitz, tensor networks/it-from-qubit, and semiclassical/stochastic gravity) and the Group O/P cross-cutting synthesis in [z.md](#); the typed-space discipline, gate list (G0–G10), and self-audit in [v6_3_1.md](#); and the compelling-layer analogy discipline in [writeup_v1_1.md](#). Nothing in Sections 4–5 promotes a claim past where its source document left it — this document only reorganizes existing debt around the gates it's owed against, so the gaps are easier to see in one place.